

RFID Hybrid Enclosure – Mechanical Design Portfolio

Balaji Jegan Maheshwaran T

Senior Mechanical Design Engineer

◆ Project Summary

Designed and developed an industrial RFID hybrid enclosure integrating plastic injection moulded housing, aluminium structural components, and CFRP PCB support plate. Responsible for complete end-to-end mechanical development from concept modelling to production-ready drawings and assembly documentation.

◆ Scope of Work

- 3D modelling of enclosure, structural plate, and mounting clamps
 - Injection moulding design with 2° draft consideration
 - Sheet metal clamp design with bend radius validation
 - CNC machined aluminium support plate design
 - PCB fitment validation and tolerance consideration
 - Creation of detailed manufacturing drawings with GD&T
 - Exploded assembly view and structured BOM documentation
 - Mass property validation using assigned material densities
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◆ Manufacturing Processes Considered

- Plastic Injection Moulding (ABS enclosure)
 - CNC Machining (Aluminium plate)
 - CNC Routing / Waterjet (CFRP plate)
 - Sheet Metal Bending (Aluminium clamp)
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◆ Key Design Considerations

- Uniform wall thickness for moldability
- Draft angle for manufacturability
- Minimum bend radius compliance
- Deburring and finishing specifications
- Fastener standardization (SS304)
- Realistic tolerance application (± 0.2 mm)

